

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2026
Parallel & Distributed Computing (CS 429)
Assignment # 2

Semester: 7th
Announced Date: 12-May-26
Total Marks: 3
Instructor Name: Dr. Zaheen Fatima
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Batch: 2022F
Due Date: 20-May-2026
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 1	Describe the performance measurement, information exchange techniques and interconnection networks in Parallel Computing.	PLO_2 (Knowledge for Computing Problem)	C2 (Understanding)

Question 1: A company is designing a data analysis application to process large datasets. Tests show that 75% of the application is parallelizable, and the rest must execute sequentially. The team wants a speedup of at least 3× to meet performance targets.

- a) How many processors are required to achieve this speedup?
- b) If only 4 processors are used, what speedup can actually be achieved?

Question 2: A climate simulation application is running on a distributed parallel computing system consisting of 16 processors, where each processor has its own local memory. The processors are connected using a mesh interconnection network.

The application divides a large dataset into equal parts, and each processor performs computations on its portion. After each computation step, processors must exchange boundary data with their neighboring processors. At the end of execution, all partial results are sent to a master processor, which computes the final result.

However, the system is experiencing:

- High communication delay
- Reduced efficiency as the number of processors increases

Tasks:

1. Identify the **parallel computing model** used in this system
2. Explain how **MPI is used for communication** in this scenario
3. Identify the **type of interconnection network** and explain its working
4. Discuss the **information exchange technique** used
5. Analyze the **performance issues (latency, efficiency)**
6. Suggest **improvements** to enhance performance

Answers

Question 1: Amdahl’s Law Application

Given: Parallelizable fraction P = 0.75, Sequential fraction S = 0.25, Target speedup >= 3x

Formula (Amdahl’s Law): Speedup(N) = 1 / (S + P/N) = 1 / (0.25 + 0.75/N)

a) Processors Required for Speedup >= 3x

Setting Speedup = 3: $3 = 1 / (0.25 + 0.75/N)$

$0.25 + 0.75/N = 1/3 \Rightarrow 0.75/N = 1/3 - 0.25 = 0.0833 \Rightarrow N = 0.75 / 0.0833 = 9$

Result: At least 9 processors are required to achieve a speedup of 3x. (Note: Maximum theoretical speedup is capped at 4x due to the 25% sequential portion.)

b) Speedup with 4 Processors

$\text{Speedup}(4) = 1 / (0.25 + 0.75/4) = 1 / (0.25 + 0.1875) = 1 / 0.4375 = 2.29$

Result: With 4 processors, the actual speedup achieved is approximately 2.29x, which does not meet the 3x performance target.

Question 2: Climate Simulation on Distributed System

i) Parallel Computing Model

This system uses the **Distributed Memory Model** (Message Passing Model). Each of the 16 processors has its own local memory and cannot directly access another processor's memory. All communication happens explicitly by passing messages. This also fits the **MIMD (Multiple Instruction, Multiple Data)** classification in Flynn's taxonomy.

ii) How MPI is Used for Communication

MPI (Message Passing Interface) handles all inter-processor communication: **MPI_Scatter** distributes the dataset to all processors; **MPI_Send / MPI_Recv (or MPI_Sendrecv)** handles boundary data exchange with neighbors after each computation step; **MPI_Gather / MPI_Reduce** collects partial results at the master processor; and **MPI_Barrier** synchronizes all processors before the next computation step.

iii) Interconnection Network: 2D Mesh

The processors are connected using a **2D Mesh Network**. The 16 processors are arranged in a 4x4 grid, where each processor connects to up to 4 immediate neighbors (North, South, East, West). Data travels hop-by-hop through intermediate nodes. The diameter (maximum hops between any two nodes) is $2(\sqrt{N} - 1) = 6$ hops for 16 nodes. The limited bisection bandwidth can cause bottlenecks during all-to-one communication (gather to master).

iv) Information Exchange Technique

Two techniques are used: (1) **Nearest-Neighbor Communication (Boundary Exchange)** — each processor shares boundary data with adjacent processors after each computation step (structured, local exchange). (2) **Collective Communication (Reduction/Gather)** — at the end, all processors send results to the master processor via MPI_Gather (many-to-one collective operation).

v) Performance Issues Analysis

High Communication Latency: Mesh topology requires multiple hops between non-adjacent processors. Boundary exchanges occur every computation step, creating frequent synchronization overhead.

Reduced Efficiency: As the number of processors N increases, communication overhead grows while computation per processor shrinks, worsening the communication-to-computation ratio. The single master processor also becomes a serial bottleneck during the final gather, and all processors must wait at barriers for the slowest processor (straggler effect).

vi) Suggested Improvements

1. Tree-based Reduction: Replace the flat gather with a tree-based reduction to reduce gather time from $O(N)$ to $O(\log N)$.

2. Non-blocking Communication: Use MPI_Isend and MPI_Irecv to overlap communication with computation of interior data.

3. Better Network Topology: Switch to a Torus network (mesh with wrap-around edges) or Fat-Tree topology to reduce diameter and balance traffic better.

4. Increase Computation Granularity: Assign larger chunks per processor to improve the computation-to-communication ratio.

5. Load Balancing: Ensure all processors receive equal workloads to avoid straggler delays at synchronization barriers.